

CFD HW#2 기계공학과 권혁순 (20241109)

1. Consider the same problem that was used in class when illustrating the method of weighted residuals. Solve the problem using the three-term approximation and compare them with the exact solution.

In the class, we have considered the following differential equation,

$$L(\phi) = \frac{d^2\phi}{dx^2} + Q = 0, \quad Q = \begin{cases} 1 & \text{at } 0 \leq x \leq \frac{l}{2} \\ 0 & \text{at } \frac{l}{2} < x \leq l \end{cases} \quad \text{B.C.s } \phi(0) = \phi(l) = 0$$

1) Exact solution

$$\frac{d^2\phi}{dx^2} = -Q \xrightarrow{\int} \frac{d\phi}{dx} = -Qx + C_* \xrightarrow{\int} \phi(x) = -\frac{1}{2}Qx^2 + C_*x + C_{**} \quad (C_*, C_{**} : \text{constant})$$

i) $0 \leq x \leq \frac{l}{2}$

$$\phi(x) = -\frac{1}{2}x^2 + C_1x + C_2$$

$$\phi(0) = 0 = C_2 \rightarrow \phi(x) = -\frac{1}{2}x^2 + C_1x$$

ii) $\frac{l}{2} < x \leq l$

$$\phi(x) = C_3x + C_4$$

$$\phi(l) = 0 = C_3l + C_4 \rightarrow \phi(x) = C_3x - C_3l$$

$$\text{At } x = \frac{l}{2}, \quad \phi\left(\frac{l}{2}\right) = -\frac{l^2}{8} + C_1 \cdot \frac{l}{2} = -C_3 \cdot \frac{l}{2} \quad \left\{ \begin{array}{l} \left(\frac{d\phi}{dx}\right)_{x=\frac{l}{2}} = -\frac{l}{2} + C_1 = C_3 \end{array} \right.$$

$$-\frac{l^2}{8} + \frac{C_1l}{2} = -\left(-\frac{l}{2} + C_1\right) \cdot \frac{l}{2} \rightarrow \therefore C_1 = \frac{3l}{8} \rightarrow C_3 = -\frac{l}{8}$$

$$\therefore \phi(x) = \begin{cases} -\frac{1}{2}x^2 + \frac{3l}{8}x, & \text{if } 0 \leq x < \frac{l}{2} \\ -\frac{l}{8}(x-l), & \text{if } \frac{l}{2} \leq x \leq l \end{cases}$$

2) Approximate solution

Let's assume an approximate solution ϕ_n that satisfies B.C.s.

$$\phi_n(x) = \sum_{i=1}^n a_i \psi_i(x) \quad (\psi_i : \text{trial function}), \quad \psi_i(0) = 0 \text{ and } \psi_i(l) = 0$$

We wish to make the residual, $R(x) = \phi_n'' + Q = 0$, small in some sense.

So, let's say $\int_0^l (\phi_n'' + Q)w_j dx = 0$ ($w(x)$: weighting function, $R \perp w$ in integral sense)

By applying integration by parts,

$$\int_0^l (\phi_n'' + Q)w_j dx = [\phi_n' w_j]_0^l - \int_0^l \phi_n' w_j' dx + \int_0^l Q w_j dx = \int_0^l (-\phi_n' w_j' + Q w_j) dx + \phi_n'(l)w_j(l) - \phi_n'(0)w_j(0)$$

2-1) In Galerkin's method, the following conditions must be satisfied.

$$\int_0^L (\phi_n'' + Q) w_j dx = 0 \quad (w_j : \text{weighting function, } j = 1 \sim n) \quad , \quad w_j(x) = \psi_j(x)$$

By applying integration by parts,

$$\int_0^L \phi_n'' w_j dx = \left[\phi_n' w_j \right]_0^L - \int_0^L \phi_n' w_j' dx \xrightarrow{\because w_j(L) = w_j(0) = 0} \int_0^L (\phi_n'' + Q) w_j dx = \int_0^L [-\phi_n' w_j' + Q w_j] dx$$

Let's solve approximately by using n -term approximation.

$$\phi_n = \sum_{i=1}^n a_i \psi_i$$

$$\text{Set } w_1 = \psi_1, w_2 = \psi_2, \dots \text{ and } w_n = \psi_n \quad (\psi_j = \sin \frac{j\pi x}{L})$$

ϕ_n satisfies the B.C.s

The equation has changed as follows.

$$\int_0^L (\phi_n'' + Q) w_j dx = \int_0^L [-\phi_n' w_j' + Q w_j] dx = -\int_0^L \left(\sum_{i=1}^n a_i \psi_i' \right) \psi_j' dx + \int_0^L Q \psi_j dx$$

We first consider the term that includes $\phi_n' w_j'$.

$$\begin{aligned} \phi_n' &= \frac{d\phi_n}{dx} = \sum_{i=1}^n a_i \psi_i' = \sum_{i=1}^n a_i \cdot \frac{d}{dx} \left(\sin \frac{i\pi x}{L} \right) = \sum_{i=1}^n a_i \cdot \frac{i\pi}{L} \cos \frac{i\pi x}{L} \\ \longrightarrow - \int_0^L \frac{d\phi_n}{dx} \cdot \frac{d\psi_j}{dx} dx &= - \int_0^L \left(\sum_{i=1}^n a_i \cdot \frac{i\pi}{L} \cos \frac{i\pi x}{L} \right) \cdot \left(\frac{j\pi}{L} \cos \frac{j\pi x}{L} \right) dx \\ &= - \sum_{i=1}^n a_i \frac{j^2 \pi^2}{L^2} \int_0^L \cos \frac{i\pi x}{L} \cdot \cos \frac{j\pi x}{L} dx \end{aligned}$$

Using the orthogonality property of trigonometric functions, we obtain

$$\int_0^L \cos \frac{i\pi x}{L} \cos \frac{j\pi x}{L} dx = \begin{cases} \frac{L}{2} & (\text{if } i=j) \\ 0 & (\text{otherwise}) \end{cases} \longrightarrow - \int_0^L \phi_n' \psi_j' dx = -a_j \cdot \left(\frac{j\pi}{L} \right)^2 \cdot \frac{L}{2} = -\frac{a_j j^2 \pi^2}{2L}$$

Next, we consider the term that includes Q .

$$\int_0^L Q \cdot \psi_j dx = \int_0^{L/2} \sin \frac{j\pi x}{L} dx + 0 = -\frac{L}{j\pi} \left[\cos \frac{j\pi x}{L} \right]_0^{L/2} = \frac{L}{j\pi} (1 - \cos \frac{j\pi}{2}) \quad \left(\because Q = \begin{cases} 1 & (0 \leq x \leq L/2) \\ 0 & (L/2 < x \leq L) \end{cases} \right)$$

Therefore, the entire equation can be written as follows and we can obtain a_j .

$$\int_0^L (\phi_n'' + Q) w_j dx = -\frac{a_j j^2 \pi^2}{2L} + \frac{L}{j\pi} (1 - \cos \frac{j\pi}{2}) = 0 \longrightarrow a_j = \frac{2L^2}{j^3 \pi^3} (1 - \cos \frac{j\pi}{2})$$

Now, we have to consider 3-term approximation to obtain approximate solution.

Since we already know the coefficient of ϕ_n , the solution derived by the Galerkin method is as follows:

$$j=1 : a_1 = \frac{2L^2}{\pi^3}$$

$$j=2 : a_2 = \frac{2L^2}{8\pi^3} \cdot 2 = \frac{L^2}{2\pi^3}$$

$$j=3 : a_3 = \frac{2L^2}{27\pi^3}$$

$$\longrightarrow \phi_3^* = a_1 \psi_1 + a_2 \psi_2 + a_3 \psi_3 = \frac{2L^2}{\pi^3} \sin \frac{\pi x}{L} + \frac{L^2}{2\pi^3} \sin \frac{2\pi x}{L} + \frac{2L^2}{27\pi^3} \sin \frac{3\pi x}{L}$$

2-2) Point collocation method

Let's select $w_1 = \delta(x - \frac{l}{4})$, $w_2 = \delta(x - \frac{l}{2})$, and $w_3 = \delta(x - \frac{3l}{4})$ for point collocation method.

Using the property of Dirac delta function, the weighted residual form takes the following form. ($\psi_j = \sin \frac{j\pi x}{l}$)

$$\int_0^l (\phi_3'' + Q) w_j dx = 0 \rightarrow \phi_3'' + Q \Big|_{x=l/4} = 0, \phi_3'' + Q \Big|_{x=l/2} = 0, \phi_3'' + Q \Big|_{x=3l/4} = 0$$

From the above three equations, all coefficients can be determined.

$$\phi_3 = \sum_{i=1}^3 a_i \psi_i = a_1 \sin \frac{\pi x}{l} + a_2 \sin \frac{2\pi x}{l} + a_3 \sin \frac{3\pi x}{l}$$

$$\phi_3'' = -a_1 \frac{\pi^2}{l^2} \sin \frac{\pi x}{l} - a_2 \frac{4\pi^2}{l^2} \sin \frac{2\pi x}{l} - a_3 \frac{9\pi^2}{l^2} \sin \frac{3\pi x}{l}$$

$$Q = \begin{cases} 1 & (0 \leq x \leq \frac{l}{2}) \\ 0 & (\frac{l}{2} < x < l) \end{cases}$$

$$\begin{aligned} \phi_3'' + Q \Big|_{x=l/4} &= -a_1 \frac{\pi^2}{l^2} \sin \frac{\pi}{4} - a_2 \frac{4\pi^2}{l^2} \sin \frac{\pi}{2} - a_3 \frac{9\pi^2}{l^2} \sin \frac{3\pi}{4} + 1 = 0 \\ &= -\frac{\sqrt{2}}{2} \cdot \frac{\pi^2}{l^2} a_1 - \frac{4\pi^2}{l^2} a_2 - \frac{\sqrt{2}}{2} \cdot \frac{9\pi^2}{l^2} a_3 + 1 = 0 \end{aligned} \quad \dots 1$$

$$\begin{aligned} \phi_3'' + Q \Big|_{x=l/2} &= -a_1 \frac{\pi^2}{l^2} \sin \frac{\pi}{2} - a_2 \frac{4\pi^2}{l^2} \sin \pi - a_3 \frac{9\pi^2}{l^2} \sin \frac{3\pi}{2} + 1 = 0 \\ &= -\frac{\pi^2}{l^2} a_1 + \frac{9\pi^2}{l^2} a_3 + 1 = 0 \end{aligned} \quad \dots 2$$

$$\begin{aligned} \phi_3'' + Q \Big|_{x=3l/4} &= -a_1 \frac{\pi^2}{l^2} \sin \frac{3\pi}{4} - a_2 \frac{4\pi^2}{l^2} \sin \frac{3\pi}{2} - a_3 \frac{9\pi^2}{l^2} \sin \frac{9\pi}{4} = 0 \\ &= -\frac{\sqrt{2}}{2} \cdot \frac{\pi^2}{l^2} a_1 + \frac{4\pi^2}{l^2} a_2 - \frac{\sqrt{2}}{2} \cdot \frac{9\pi^2}{l^2} a_3 = 0 \end{aligned} \quad \dots 3$$

$$\textcircled{2} - \textcircled{1} \Rightarrow \frac{8\pi^2}{l^2} a_2 - 1 = 0 \rightarrow a_2 = \frac{l^2}{8\pi^2}$$

$$\textcircled{1}, \textcircled{3} \Rightarrow -\frac{\sqrt{2}}{2} \cdot \frac{\pi^2}{l^2} a_1 - \frac{\sqrt{2}}{2} \cdot \frac{9\pi^2}{l^2} a_3 + \frac{1}{2} = 0$$

$$\frac{\sqrt{2}}{2} \times \textcircled{2} \Rightarrow -\frac{\sqrt{2}}{2} \cdot \frac{\pi^2}{l^2} a_1 + \frac{\sqrt{2}}{2} \cdot \frac{9\pi^2}{l^2} a_3 + \frac{\sqrt{2}}{2} = 0$$

$$-\sqrt{2} \cdot \frac{9\pi^2}{l^2} a_3 + \frac{1-\sqrt{2}}{2} = 0 \rightarrow a_3 = \frac{1-\sqrt{2}}{2\sqrt{2}} \cdot \frac{l^2}{9\pi^2} = \frac{(\sqrt{2}-2)l^2}{36\pi^2}$$

$$\textcircled{2} \rightarrow -\frac{\pi^2}{l^2} a_1 + \frac{1+\sqrt{2}}{2\sqrt{2}} + 1 = 0 \rightarrow a_1 = \frac{1+\sqrt{2}}{2\sqrt{2}} \cdot \frac{l^2}{\pi^2} = \frac{(2+\sqrt{2})l^2}{4\pi^2}$$

$$\therefore \phi_3^p = a_1 \sin \frac{\pi x}{l} + a_2 \sin \frac{2\pi x}{l} + a_3 \sin \frac{3\pi x}{l} = \frac{(2+\sqrt{2})l^2}{4\pi^2} \cdot \sin \frac{\pi x}{l} + \frac{l^2}{8\pi^2} \sin \frac{2\pi x}{l} + \frac{(\sqrt{2}-2)l^2}{36\pi^2} \cdot \sin \frac{3\pi x}{l}$$

2-3) Subdomain collocation method

Let's set

$$w_1 = \begin{cases} 1 & (0 \leq x \leq \frac{l}{3}) \\ 0 & (\frac{l}{3} < x < l) \end{cases}, \quad w_2 = \begin{cases} 1 & (\frac{l}{3} \leq x \leq \frac{2l}{3}) \\ 0 & (\text{otherwise}) \end{cases}, \quad w_3 = \begin{cases} 1 & (\frac{2l}{3} \leq x \leq l) \\ 0 & (\text{otherwise}) \end{cases}$$

$$\int_0^l (\phi_s'' + Q) w_j dx = 0$$

$$(\psi_j = \sin \frac{j\pi x}{l})$$

$$j = 1 \rightarrow \int_0^{l/3} (a_1 \psi_1'' + a_2 \psi_2'' + a_3 \psi_3'' + 1) dx = 0 \quad \dots \textcircled{1}$$

$$j = 2 \rightarrow \int_{l/3}^{2l/3} (a_1 \psi_1'' + a_2 \psi_2'' + a_3 \psi_3'' + 1) dx + \int_{l/3}^{2l/3} (a_1 \psi_1'' + a_2 \psi_2'' + a_3 \psi_3'') dx = 0 \quad \dots \textcircled{2}$$

$$j = 3 \rightarrow \int_{2l/3}^l (a_1 \psi_1'' + a_2 \psi_2'' + a_3 \psi_3'') dx = 0 \quad \dots \textcircled{3}$$

$$\textcircled{1} : \int_0^{l/3} \left[a_1 \cdot \left(-\frac{\pi^2}{l^2} \right) \sin \frac{\pi x}{l} + a_2 \cdot \left(-\frac{4\pi^2}{l^2} \right) \sin \frac{2\pi x}{l} + a_3 \cdot \left(-\frac{9\pi^2}{l^2} \right) \sin \frac{3\pi x}{l} + 1 \right] dx = 0$$

$$\rightarrow a_1 \left(-\frac{\pi^2}{l^2} \right) \left(-\frac{l}{\pi} \right) \left(\frac{1}{2} - 1 \right) + a_2 \left(-\frac{4\pi^2}{l^2} \right) \left(-\frac{l}{2\pi} \right) \left(-\frac{1}{2} - 1 \right) + a_3 \left(-\frac{9\pi^2}{l^2} \right) \left(-\frac{l}{3\pi} \right) (-1 - 1) + \frac{l}{3} = 0$$

$$= -\frac{\pi}{2l} a_1 - \frac{3\pi}{l} a_2 - \frac{6\pi}{l} a_3 + \frac{l}{3} = 0 \quad \dots \textcircled{3}$$

$$\textcircled{2} : \int_{l/3}^{2l/3} \left[a_1 \cdot \left(-\frac{\pi^2}{l^2} \right) \sin \frac{\pi x}{l} + a_2 \cdot \left(-\frac{4\pi^2}{l^2} \right) \sin \frac{2\pi x}{l} + a_3 \cdot \left(-\frac{9\pi^2}{l^2} \right) \sin \frac{3\pi x}{l} + 1 \right] dx$$

$$+ \int_{l/3}^{2l/3} \left[a_1 \cdot \left(-\frac{\pi^2}{l^2} \right) \sin \frac{\pi x}{l} + a_2 \cdot \left(-\frac{4\pi^2}{l^2} \right) \sin \frac{2\pi x}{l} + a_3 \cdot \left(-\frac{9\pi^2}{l^2} \right) \sin \frac{3\pi x}{l} \right] dx = 0$$

$$\rightarrow a_1 \left(-\frac{\pi^2}{l^2} \right) \left(-\frac{l}{\pi} \right) \left(0 - \frac{1}{2} \right) + a_2 \left(-\frac{4\pi^2}{l^2} \right) \left(-\frac{l}{2\pi} \right) \left(-1 - \left(-\frac{1}{2} \right) \right) + a_3 \left(-\frac{9\pi^2}{l^2} \right) \left(-\frac{l}{3\pi} \right) (0 - (-1)) + \left(\frac{l}{2} - \frac{l}{3} \right)$$

$$+ a_1 \left(-\frac{\pi^2}{l^2} \right) \left(-\frac{l}{\pi} \right) \left(-\frac{1}{2} - 0 \right) + a_2 \left(-\frac{4\pi^2}{l^2} \right) \left(-\frac{l}{2\pi} \right) \left(-\frac{1}{2} - (-1) \right) + a_3 \left(-\frac{9\pi^2}{l^2} \right) \left(-\frac{l}{3\pi} \right) (1 - 0) = 0$$

$$= -\frac{\pi}{l} a_1 + \frac{6\pi}{l} a_3 + \frac{l}{6} = 0 \quad \dots \textcircled{4}$$

$$\textcircled{3} : \int_{2l/3}^l \left[a_1 \cdot \left(-\frac{\pi^2}{l^2} \right) \sin \frac{\pi x}{l} + a_2 \cdot \left(-\frac{4\pi^2}{l^2} \right) \sin \frac{2\pi x}{l} + a_3 \cdot \left(-\frac{9\pi^2}{l^2} \right) \sin \frac{3\pi x}{l} \right] dx = 0$$

$$\rightarrow a_1 \left(-\frac{\pi^2}{l^2} \right) \left(-\frac{l}{\pi} \right) \left(-1 - \left(-\frac{1}{2} \right) \right) + a_2 \left(-\frac{4\pi^2}{l^2} \right) \left(-\frac{l}{2\pi} \right) \left(1 - \left(-\frac{1}{2} \right) \right) + a_3 \left(-\frac{9\pi^2}{l^2} \right) \left(-\frac{l}{3\pi} \right) ((-1) - 1) = 0$$

$$= -\frac{\pi}{2l} a_1 + \frac{3\pi}{l} a_2 - \frac{6\pi}{l} a_3 = 0 \quad \dots \textcircled{5}$$

$$\left. \begin{array}{l} \textcircled{3} + \textcircled{5} \rightarrow -\frac{\pi}{l} a_1 - \frac{12\pi}{l} a_3 + \frac{l}{3} = 0 \\ \textcircled{4} \rightarrow -\frac{\pi}{l} a_1 + \frac{6\pi}{l} a_3 + \frac{l}{6} = 0 \end{array} \right\}$$

$$-\frac{18\pi}{l} a_3 + \frac{l}{6} = 0 \rightarrow a_3 = \frac{l^2}{108\pi}, \quad a_1 = \frac{2l^2}{9\pi} \rightarrow a_2 = \frac{l^2}{18\pi}$$

$$\therefore \phi_s = a_1 \sin \frac{\pi x}{l} + a_2 \sin \frac{2\pi x}{l} + a_3 \sin \frac{3\pi x}{l} = \frac{2l^2}{9\pi} \sin \frac{\pi x}{l} + \frac{l^2}{18\pi} \sin \frac{2\pi x}{l} + \frac{l^2}{108\pi} \sin \frac{3\pi x}{l}$$

The analytic and approximate solutions we obtained are as follows.

* Exact solution

$$\therefore \phi(x) = \begin{cases} -\frac{1}{2}x^2 + \frac{3l}{8}x, & \text{if } 0 \leq x < \frac{l}{2} \\ -\frac{l}{8}(x-l), & \text{if } \frac{l}{2} \leq x \leq l \end{cases}$$

* Galerkin method

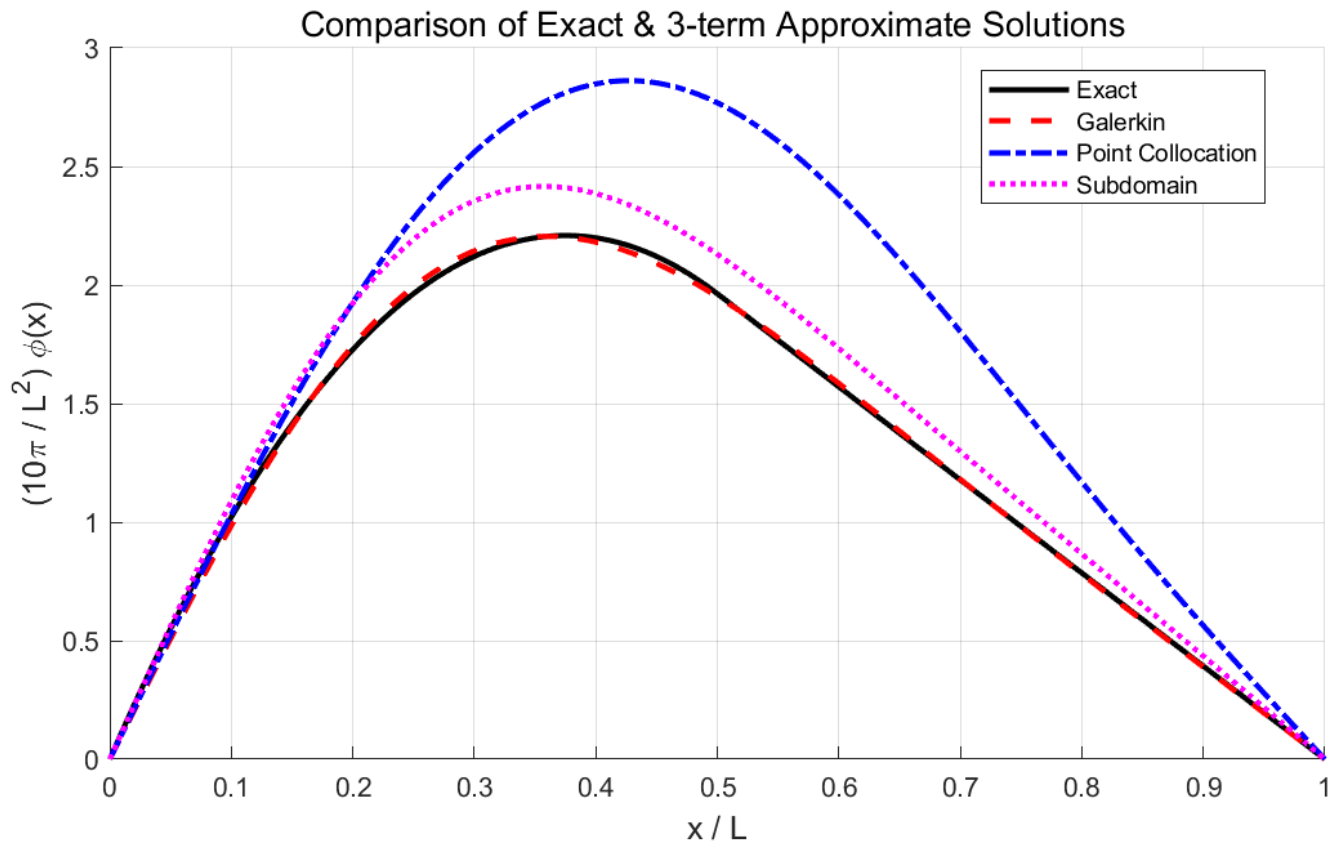
$$\phi_3 = \frac{2l^2}{\pi^3} \sin \frac{\pi x}{l} + \frac{l^2}{2\pi^3} \sin \frac{2\pi x}{l} + \frac{2l^2}{27\pi^3} \sin \frac{3\pi x}{l}$$

* Point collocation method

$$\phi_3 = \frac{(2+\sqrt{2})l^2}{4\pi^2} \cdot \sin \frac{\pi x}{l} + \frac{l^2}{8\pi^2} \sin \frac{2\pi x}{l} + \frac{(\sqrt{2}-2)l^2}{36\pi^2} \cdot \sin \frac{3\pi x}{l}$$

* Subdomain collocation method

$$\phi_3 = \frac{2l^2}{9\pi} \sin \frac{\pi x}{l} + \frac{l^2}{18\pi} \sin \frac{2\pi x}{l} + \frac{l^2}{108\pi} \sin \frac{3\pi x}{l}$$



The above plot compares the exact solution to three approximate solutions.

All three 3-term approximations capture the overall shape of the exact solution.

The Galerkin method tracks the exact curve very well, while the other methods show slight deviations in the peak region.

As shown in the figure above, the Galerkin method produces the best approximation to the exact solution, followed by the Subdomain collocation method and Point collocation method, respectively.

2. Consider the ordinary differential equation

$$\frac{d^2\phi}{dx^2} + 10x^2 = 0,$$

subject to the boundary conditions

$$\phi(0) = 0 \quad \phi(1) = 1$$

a. Obtain the analytical solution for ϕ as a function of x .

$$\frac{d^2\phi}{dx^2} = -10x^2$$

By taking integration,

$$\frac{d\phi}{dx} = -\frac{10}{3}x^3 + C_1 \longrightarrow \phi(x) = -\frac{5}{6}x^4 + C_1x + C_2 \quad (C_1, C_2 : \text{constant})$$

Applying the boundary conditions,

$$\begin{cases} \phi(0) = C_2 = 0 \longrightarrow \therefore C_2 = 0 \\ \phi(1) = -\frac{5}{6} + C_1 = 1 \longrightarrow \therefore C_1 = \frac{11}{6} \end{cases}$$

$$\therefore \phi(x) = -\frac{5}{6}x^4 + \frac{11}{6}x$$

b. To obtain an approximate solution assume a solution of the following form

$$\phi = x + \varphi$$

Then

$$\frac{d^2\varphi}{dx^2} + 10x^2 = 0$$

subject to

$$\varphi(0) = 0 \quad \varphi(1) = 0$$

Considering the two-term approximation of the following form

$$\varphi \approx \varphi_2 = \sum a_i \sin i\pi x; i = 1 \text{ and } 2,$$

solve for $\varphi_2(x)$ using the Galerikin, point collocation, and subdomain methods.

Applying 2-term approximation,

$$\varphi_2(x) = a_1 \sin \pi x + a_2 \sin 2\pi x$$

$$\varphi_2''(x) = -a_1 \pi^2 \sin \pi x - a_2 \cdot (4\pi^2) \cdot \sin 2\pi x$$

Then, the ODE is as follows.

$$\frac{d^2\varphi}{dx^2} + 10x^2 = 0 \longrightarrow -a_1 \pi^2 \sin \pi x - a_2 \cdot (4\pi^2) \cdot \sin 2\pi x + 10x^2 = 0$$

i) Galerkin method

From the problem statement, we can determine the weighting function as follows.

$$\varphi_2(x) = \sum_{i=1}^2 a_i (\sin i\pi x) \rightarrow w_j = \sin j\pi x \quad (j = 1, 2)$$

$$\int_0^1 (\varphi'' + 10x^2) w_j dx = 0$$

Applying integration by parts, the term that includes φ_2'' takes the following form.

$$\int_0^1 \varphi_2'' w_j dx \rightarrow \varphi_2'(1) w_j(1) - \varphi_2'(0) w_j(0) - \int_0^1 \varphi_2' w_j' dx = \int_0^1 \left[- \left(\sum_{i=1}^2 a_i \cdot (i\pi) \cos i\pi x \right) \cdot (j\pi) \cos j\pi x \right] dx = 0$$

$\because w_j(0) = w_j(1) = 0$

Using the orthogonality property of trigonometric functions, we obtain

$$\int_0^1 \cos i\pi x \cdot \cos j\pi x dx = \begin{cases} \frac{1}{2} & (\text{if } i=j) \\ 0 & (\text{otherwise}) \end{cases} \rightarrow - \int_0^1 \varphi_2' w_j' dx = -a_j \cdot (j\pi)^2 \cdot \frac{1}{2} = -\frac{a_j j^2 \pi^2}{2}$$

The term that includes " $10x^2$ " takes the following form.

$$\begin{aligned} \int_0^1 10 \cdot x^2 \cdot \sin j\pi x dx &= 10 \left[x^2 \cdot \left(\frac{-1}{j\pi} \right) \cos j\pi x \right]_0^1 - 10 \int_0^1 \left(\frac{-1}{j\pi} \right) \cos j\pi x \cdot (2x) dx \\ &= 10 \left(\frac{-1}{j\pi} \right) \cos j\pi + \left(\frac{20}{j\pi} \right) \left\{ \left[x \cdot \frac{1}{j\pi} \cdot \sin j\pi x \right]_0^1 - \int_0^1 \left(\frac{1}{j\pi} \right) \sin j\pi x dx \right\} \\ &= 10 \left(\frac{-1}{j\pi} \right) \cos j\pi + \left(\frac{20}{j\pi} \right) \left(\frac{1}{j\pi} \right)^2 (\cos j\pi - 1) \\ &= \frac{10}{j\pi} \left(\frac{2}{(j\pi)^2} - 1 \right) \cos j\pi - \frac{20}{(j\pi)^3} \end{aligned}$$

Therefore, the entire equation can be written as follows and we can obtain a_j .

$$\int_0^1 (\varphi'' + 10x^2) w_j dx = -\frac{a_j j^2 \pi^2}{2} + \frac{10}{j\pi} \left(\frac{2}{(j\pi)^2} - 1 \right) \cos j\pi - \frac{20}{(j\pi)^3} = 0$$

$$\text{for } j=1, \quad -\frac{\pi^2}{2} a_1 + \frac{10}{\pi} \left(\frac{2}{\pi^2} - 1 \right) (-1) - \frac{20}{\pi^3} = 0 \rightarrow a_1 = \frac{20}{\pi^3} \left(1 - \frac{4}{\pi^2} \right)$$

$$\text{for } j=2, \quad -2\pi^2 a_2 + \frac{5}{\pi} \left(\frac{1}{2\pi^2} - 1 \right) - \frac{5}{2\pi^3} = 0 \rightarrow a_2 = -\frac{5}{2\pi^3}$$

The approximate solution derived by the Galerkin method is as follows.

$$\therefore \varphi_2^q(x) = \frac{20}{\pi^3} \left(1 - \frac{4}{\pi^2} \right) \sin \pi x - \frac{5}{2\pi^3} \sin 2\pi x$$

$$\therefore \phi_2^q(x) = x + \varphi_2^q(x) = x + \frac{20}{\pi^3} \left(1 - \frac{4}{\pi^2} \right) \sin \pi x - \frac{5}{2\pi^3} \sin 2\pi x$$

ii) Point collocation method

Let's select w_j as follows.

$$w_1 = \delta(x - \frac{1}{3}), w_2 = \delta(x - \frac{2}{3}) \longrightarrow \int_0^1 (\varphi_2'' + 10x^2) w_j dx = 0$$

By the property of direct-delta function,

$$\longrightarrow \varphi_2'' + 10x^2 \Big|_{x=\frac{1}{3}} = 0, \quad \varphi_2'' + 10x^2 \Big|_{x=\frac{2}{3}} = 0$$

$$x = \frac{1}{3} : -a_1 \pi^2 \sin \frac{\pi}{3} - a_2 \cdot (4\pi^2) \cdot \sin \frac{2\pi}{3} + \frac{10}{9} = 0$$

$$- \frac{\sqrt{3}\pi^2}{2} a_1 - \frac{4\sqrt{3}\pi^2}{2} a_2 + \frac{10}{9} = 0 \quad \dots 1$$

$$x = \frac{2}{3} : -a_1 \pi^2 \sin \frac{2\pi}{3} - a_2 \cdot (4\pi^2) \cdot \sin \frac{4\pi}{3} + \frac{40}{9} = 0$$

$$- \frac{\sqrt{3}\pi^2}{2} a_1 + \frac{4\sqrt{3}\pi^2}{2} a_2 + \frac{40}{9} = 0 \quad \dots 2$$

$$\textcircled{2} - \textcircled{1} \longrightarrow 4\sqrt{3}\pi^2 a_2 + \frac{10}{3} = 0 \longrightarrow a_2 = -\frac{10}{3} \cdot \frac{1}{4\sqrt{3}\pi^2} = -\frac{5\sqrt{3}}{18\pi^2}$$

$$\textcircled{1} \longrightarrow -\frac{\sqrt{3}\pi^2}{2} a_1 + \frac{5}{3} + \frac{10}{9} = 0 \longrightarrow a_1 = \frac{50\sqrt{3}}{27\pi^2}$$

$$\therefore \varphi_2^p(x) = \frac{50\sqrt{3}}{27\pi^2} \sin \pi x - \frac{5\sqrt{3}}{18\pi^2} \sin 2\pi x$$

$$\therefore \phi_2^p(x) = x + \varphi_2^p(x) = x + \frac{50\sqrt{3}}{27\pi^2} \sin \pi x - \frac{5\sqrt{3}}{18\pi^2} \sin 2\pi x$$

iii) Subdomain collocation method

Let's select w_j as follows.

$$w_1 = \begin{cases} 1 & (0 \leq x \leq \frac{1}{2}) \\ 0 & (\frac{1}{2} < x < 1) \end{cases} \quad w_2 = \begin{cases} 0 & (0 \leq x \leq \frac{1}{2}) \\ 1 & (\frac{1}{2} < x \leq 1) \end{cases}$$

$$\int_0^1 (\varphi_2'' + 10x^2) w_j dx = 0 \longrightarrow \int_0^1 [-a_1 \pi^2 \sin \pi x - a_2 \cdot (4\pi^2) \cdot \sin 2\pi x + 10x^2] w_j dx = 0$$

$$\text{for } j=1, \int_0^{1/2} [-a_1 \pi^2 \sin \pi x - a_2 \cdot (4\pi^2) \cdot \sin 2\pi x + 10x^2] dx = 0$$

$$\longrightarrow (-a_1 \pi^2) \left[\frac{-1}{\pi} \cos \pi x \right]_0^{1/2} - a_2 \cdot (4\pi^2) \left[\frac{-1}{2\pi} \cos 2\pi x \right]_0^{1/2} + 10 \cdot \left[\frac{1}{3} x^3 \right]_0^{1/2} = 0$$

$$\longrightarrow -\pi a_1 - 4\pi a_2 + \frac{5}{12} = 0 \quad \dots 1$$

$$\text{for } j=2, \int_{1/2}^1 [-a_1 \pi^2 \sin \pi x - a_2 \cdot (4\pi^2) \cdot \sin 2\pi x + 10x^2] dx = 0$$

$$\longrightarrow -\pi a_1 + 4\pi a_2 + \frac{35}{12} = 0 \quad \dots 2$$

$$\textcircled{1} - \textcircled{2} : -8\pi a_2 - \frac{5}{2} = 0 \longrightarrow a_2 = -\frac{5}{16\pi}$$

$$\textcircled{1} \longrightarrow a_1 = \frac{5}{32\pi}$$

$$\therefore \varphi_2^s(x) = \frac{5}{32\pi} \sin \pi x - \frac{5}{16\pi} \sin 2\pi x$$

$$\therefore \phi_2^s(x) = x + \frac{5}{32\pi} \sin \pi x - \frac{5}{16\pi} \sin 2\pi x$$

c. Compare graphically the approximate solution obtained in b with the analytic solution.

* Exact solution

$$\phi(x) = -\frac{5}{6}x^4 + \frac{11}{6}x$$

* Galerkin method

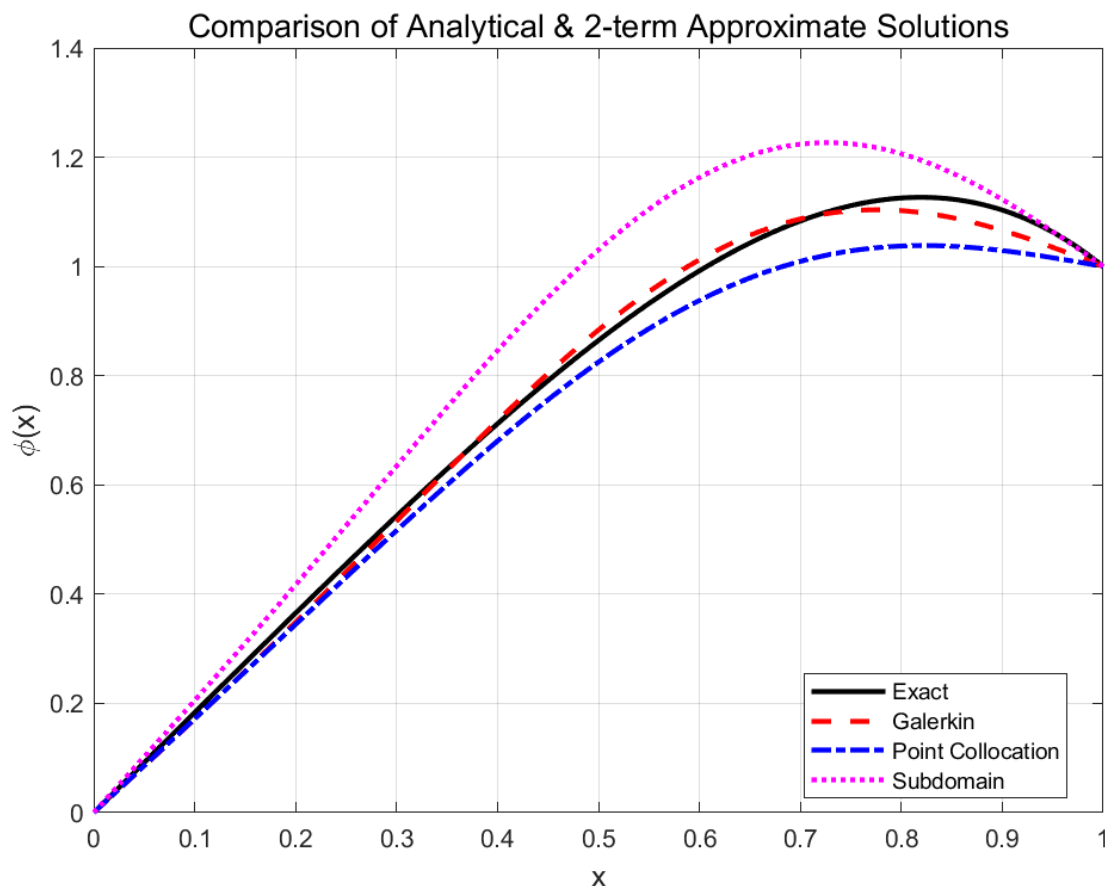
$$\phi_2^G(x) = x + \varphi_2^G(x) = x + \frac{20}{\pi^3} \left(1 - \frac{4}{\pi^2}\right) \sin \pi x - \frac{5}{2\pi^3} \sin 2\pi x$$

* Point collocation method

$$\phi_2^P(x) = x + \varphi_2^P(x) = x + \frac{50\sqrt{3}}{27\pi^2} \sin \pi x - \frac{5\sqrt{3}}{18\pi^2} \sin 2\pi x$$

* Subdomain collocation method

$$\phi_2^S(x) = x + \varphi_2^S(x) = x + \frac{5}{3\pi} \sin \pi x - \frac{5}{16\pi} \sin 2\pi x$$



All three 2-term approximations capture the overall shape of the exact (analytic) solution.

The Galerkin method aligns closest with the exact curve across the domain.

While the point collocation method approximates the exact solution quite well, noticeable errors occur near the peak.

In contrast, the subdomain method shows an overall overshooting behavior.